

Phase 1: Classical Segmentation

Raw Point Cloud

Output: $N \times 4$ array (x, y, z, r)
from .bin file

Stage 1: Noise Removal

In: raw point cloud
Out: filtered point cloud (no noise/outliers)
Method: $z > -2.0 \text{ m} + \text{stat. outlier } (k=20, \sigma=2)$

Stage 2: Voxel Downsampling

In: filtered point cloud
Out: downsampled point cloud
Method: voxel grid, size = 0.05 m

Stage 3: Ground Removal

In: downsampled point cloud
Out: object points + ground plane model $[a, b, c, d]$
Method: RANSAC ($d=0.2 \text{ m}$) + normal check

Stage 4: Clustering

In: object points (non-ground)
Out: cluster labels per point
Method: HDBSCAN (min_cluster=10, min_samples=5)

Stage 5: Geometric Filtering

In: point clusters + ground plane
Out: filtered clusters with OBBs
Method: volume $\geq 0.5 \text{ m}^3$, height $\leq 3.0 \text{ m}$

Stage 6: Classification

In: filtered clusters + OBB dimensions
Out: classified objects (label + confidence)
Method: heuristic rules on bbox extents

Multi-Frame Tracking

In: per-frame detections (global coords)
Out: persistent object IDs across frames
Method: Hungarian assignment ($d_{max}=2.0 \text{ m}$)

Phase 2: Deep Learning Completion

Stage 7: Point Cloud Completion

In: partial object point cloud (2,048 pts)
Out: completed point cloud (16,384 pts)
Method: PCN (trained on ShapeNet)